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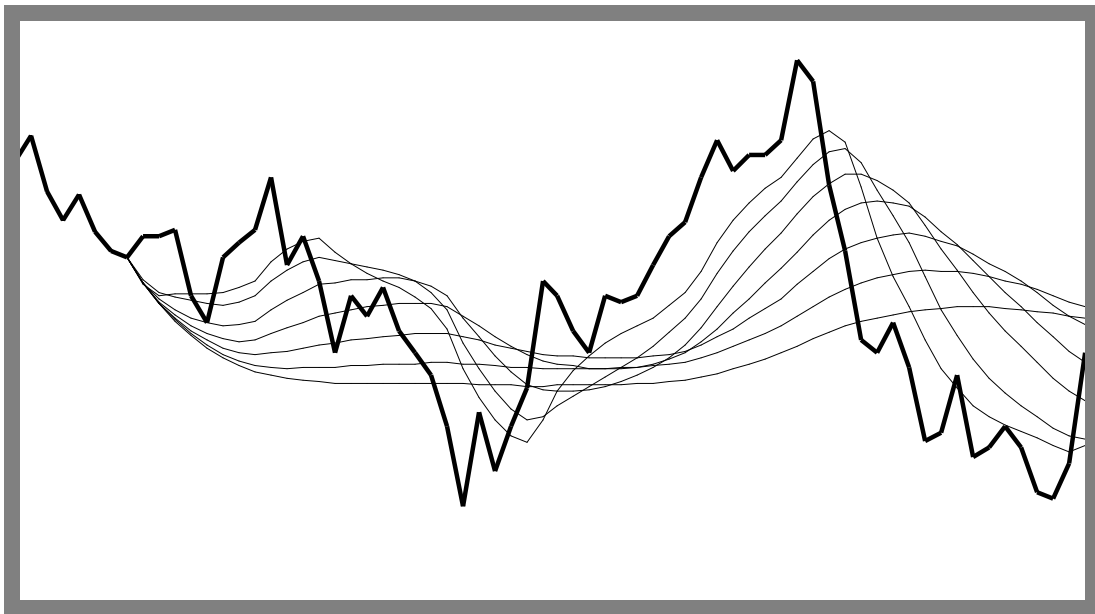
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JURIK RESEARCH

JMA

Jurik Moving Average

Add-In Tool for Omega Research Software *



USER'S GUIDE

* Omega Research TradeStation / SuperCharts 4 or TradeStation / ProSuite 2000

Installation Instructions

The software accompanying this manual is designed to be used inside Omega Research's TradeStation 4, SuperCharts 4, TradeStation 2000 or ProSuite 2000. If you do not have an up-to-date version, call Omega Research for the appropriate upgrade.

Getting Jurik's tools into an Omega Research software application is a 2-step process:

- **INSTALLATION:** to create .ELA or .ELS files containing the Jurik modules.
- **TRANSFER:** to move the modules into your Omega Research software product.

Step 1: INSTALLATION

1. The name of the installation software is JRSACT_2.EXE
2. If you **HAVE** received a password from Jurik Research (on a colored sheet of paper that came with this software), then **SKIP THIS STEP** and proceed to step #3. Otherwise, using Windows' Explorer (or File Manager), double-click or run the installer program JRSACT_2.EXE. If you have TradeStation 4 or SuperCharts 4, enter your Omega security block number if you have one. (SuperCharts end-of-day users probably have no security block number so they should leave the field blank.) If you have TradeStation 2000 or ProSuite 2000, enter your Omega Research Customer ID number. Working through, select all the Jurik tools you currently have license to use. Eventually the program will give you an **identification code**. To receive your activation password from us, e-mail to: nfs@nfsmith.net, or call 323-258-4860 or fax 323-258-0598, and tell us your name, phone number, the platform you are installing to (TS4, SC4, TS2000, PS2000) and the identification code. After receiving your password, proceed to step #3.
3. Close your Omega Research software application (SuperCharts, TradeStation or ProSuite). Leaving it open may interfere with the installation process. You do NOT need to shut down the data server.
4. Using Windows' Explorer (or File Manager), double-click or run JRSACT_2.EXE. Enter your Omega security block number or Customer ID number, if applicable. (SuperCharts end-of-day users probably have no security block number.) Also enter your password. Click OK. Select all and only those tools you currently have license to use, otherwise the password will not be accepted. Click OK. Follow other instructions on screen.
5. At this point, all the Jurik tools (eg. JMA, VEL, CFB, WAV, DDR, RSX) you currently have license to use were placed as Easy Language files on your hard drive. The next page shows how to transfer (import) them into your Omega software product.

Step 2: TRANSFER / IMPORT

Transferring to TradeStation 4.0 / SuperCharts 4.0

1. Run TOOLS / VERIFY_ALL to ensure all linkages are correct.
2. Using either QuickEditor or PowerEditor, use the FILE / OPEN command to bring up a dialog box, then press the TRANSFER button.
3. Select "**Transfer FROM Easy Language Archive File**" and press the OK button.
4. Transfer in JMA, whose default filepath is C:\JRSOMEGA\EASYLANG\JRC_JMA.ELA. (You may have elected to change this filepath during installation.) Enter the filepath and press OK.
5. Select "Transfer All" and press OK. All JMA related tools will be automatically transferred.
6. Do not assume all transferred modules were properly verified. Execute TOOLS \ VERIFY_ALL to ensure all linkages are correct.

Importing to TradeStation 2000 / ProSuite 2000

1. Using the PowerEditor, execute FILES \ VERIFY_ALL to ensure all linkages are correct.
2. Execute the FILE \ IMPORT_and_EXPORT command.
3. Select "**Import Easy Language Archive Files**" to import .ELS files. Press the OK button.
4. Import JMA, whose default filepath is C:\JRSOMEGA\EASYLANG\JMA.ELS. (You may have elected to change this filepath during installation.) Enter the filepath or use the browser to find it. Make sure you select to import all studies contained within the ELS file. Press OK.
5. During importation, it may want to load the same function several times, so you may repeatedly see a dialog box asking if you want to overwrite an already existing module. To speed up the import process, select "YES TO ALL".
6. Do not assume all the imported modules were properly verified. Execute the command FILES \ VERIFY ALL to ensure all linkages are correct.

IMPORTANT NOTICE to TradeStation/ProSuite 2000 Users

INCOMPATIBILITY

TradeStation 2000 and ProSuite 2000 (hereafter referred to as "TS2000") are 32-bit programs, which makes them completely different than TradeStation 4 and SuperCharts 4 (hereafter referred to as "TS4/SC4"), which are 16-bit programs. Consequently, the Jurik modules designed for TS4/SC4 are not compatible with TS2000. **To get Jurik modules for TS2000, you must run the installer and designate that platform.**

EXPANDED NAMES

Easy Language studies (functions, indicators and systems) will include during transfers (imports & exports) all functions required to make them work. Therefore, any studies that you import to TS2000 that were developed in TS4/SC4 will also transfer with them any Jurik functions that they utilize. Because these functions are not compatible with TS2000 (see incompatibility notice above), it is imperative that they not overwrite the Jurik modules already installed in TS2000.

To accomplish this, the names of all Jurik functions, indicators and systems for TS2000 have been expanded to include the suffix "2k". For example, if this user manual refers to a function named **JRC.JMA**, its expanded name for TS2000 is **JRC.JMA.2k**

You will need to modify any studies you transferred in from TS4/SC4 so that they will now include the ".2k" suffix on the names of all Jurik functions that they utilize.

Executive Summary

brief instructions for those who don't read user manuals

Use the **JRC JMA** indicator just as if it were an exponential moving average. It uses a proprietary user function, called **JRC.JMA**. You can code your own Easy Language modules to employ the user function as follows:

value1 = JRC.JMA (SERIES , LENGTH , PHASE)

SERIES is the series to be filtered, such as the daily closing price. To use closing prices, replace "SERIES" in the above expression with "close". This input series can also be any QuickEditor expression that produces a series. For example, "SERIES" could be replaced by "7+RSI (close, 14)".

LENGTH determines the degree of smoothness and it can be any positive value. Small values make the moving average respond rapidly to price change and larger values produce smoother, flatter curves. Typical values for LENGTH range from 5 to 80. You can even use decimal numbers, such as 28.3.

PHASE affects the amount of lag (delay). PHASE ranges from -100 (max lag) to +100 (min lag). Its default value is 0. NOTE: lower lag tends to produce larger overshoot during price gaps, so you need to consider the trade-off between lag and overshoot and select a value for PHASE that balances your trading system's needs.

There may be times when you want to dynamically alter the inputs to JMA on a bar-to-bar basis. As such, you would create new variables and feed their values to JMA. For this purpose, we have a special version of JMA, called **JRC.JMA.flex**. In the following example, all three input values are calculated just prior to feeding JMA:

```
series = close + 0.5 * stdev ( high , 10 ) ;  
length = 35 + RSI ( close , 13 ) ;  
phase = momentum ( high , 10 ) ;  
result = JRC.JMA.flex ( series , length , phase ) ;
```

NOTE: Although **JRC.JMA.flex** has this advantage over **JRC.JMA**, it also has two important disadvantages. Both are directly the result of the properties of type SIMPLE user functions in Easy Language. You need to decide which version of JMA is more appropriate in your coding. The two disadvantages of the flex version are:

- 1) **JRC.JMA.flex** does not produce a time series. Consequently, you cannot reference past values of it directly. However, you can do so indirectly by referring its output to a variable and then seeking past values of that variable. For example:

```
INVALID EXPRESSION:      result = JRC.JMA.flex ( series,length,phase)[7] ;  
VALID EXPRESSION:        value1 = JRC.JMA.flex ( series,length,phase) ;  
                           result = value1[7] ;
```

Note: This method of referencing past values of variables is not permitted inside type-SIMPLE user functions.

- 2) **JRC.JMA.flex** is not automatically evaluated on every bar. You must control when it gets evaluated. This can be advantageous. For instance, you can take a moving average of only Tuesday's closing prices when using a *daily* chart, by writing code so that JMA is called only on Tuesdays. For example:

```
if (DateOfWeek(Date) = 2) then result = JRC.JMA.flex (close,length,phase) ;
```

Max number of bars JMA will reference equals 30 plus whatever the input to JMA also requires.

INPUT to JMA	FULL EXPRESSION	MaxBarsBack
close	JRC.JMA (close,21,0)	30 + 0
close[5]	JRC.JMA (close[5],21,0)	30 + 5
RSI(close[5],14)	JRC.JMA (RSI(close[5],14), 21,0)	30 + 14 + 5

As a final example, you can build a phase enhanced MACD oscillator using JMA as follows:

```
MACD_osc = JRC.JMA(h+1, 20, -100) - JRC.JMA(h+1, 40, 0)
```

Why use JMA ?

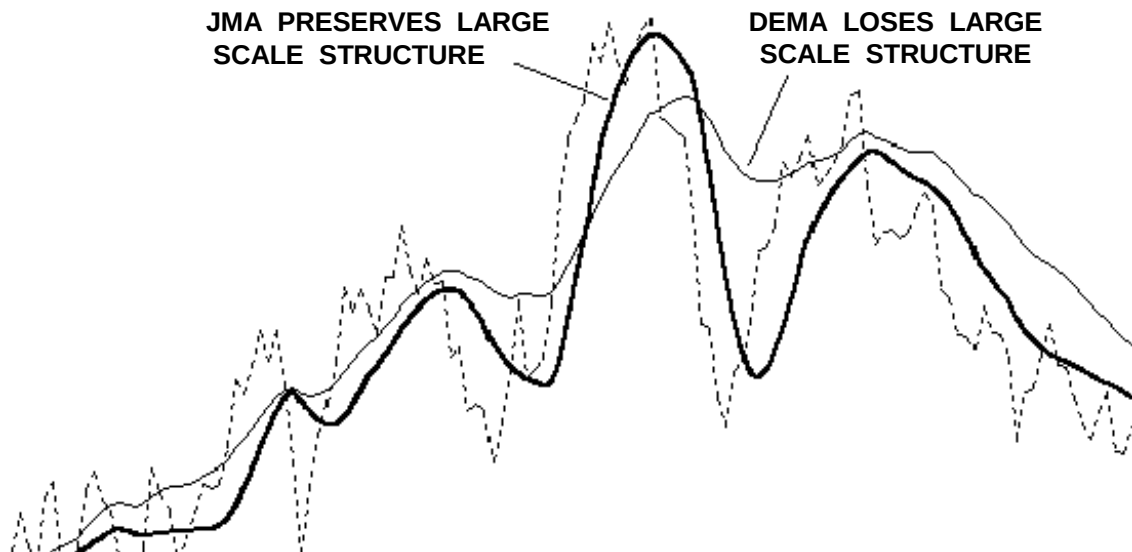
The SMART MOVING AVERAGE by Jurik Research

Daily prices produce a time series with some amount of random fluctuations. To remove this noise, market technicians typically use moving average (MA) filters. Only JMA excels in all four benchmarks of a truly great filter...

BENCHMARK #1: ACCURACY

Moving Average (MA) filters have an adjustable parameter that controls its speed. Speed governs two opposing properties of a filter: smoothness (lack of random zigzagging) and accuracy (closeness to the original data). That is, the smoother a filter becomes, the less it accurately resembles the original time series. This makes sense, since we do not want to accurately track zigzagging noise within our data.

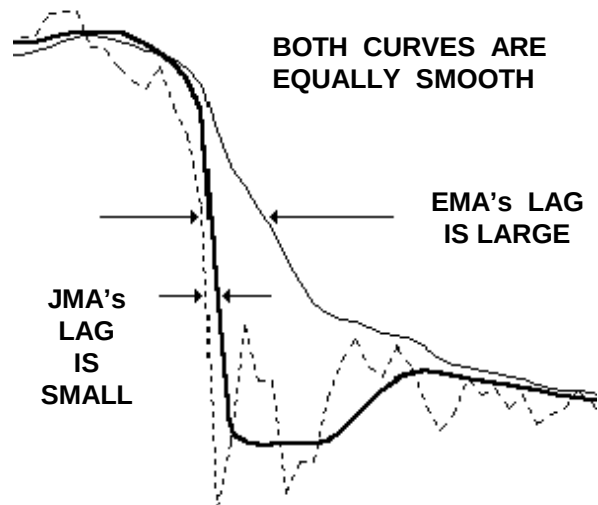
Because the financial investor tries to apply just enough smoothness to filter out noise without removing important structure in price activity. For example, in the chart below, the popular Double Exponential Moving Average (DEMA) is just as smooth as JMA yet DEMA fails to track large scale structure (the big cycles). On the other hand, JMA follows the cyclic action very well.



BENCHMARK #2: TIMELINESS

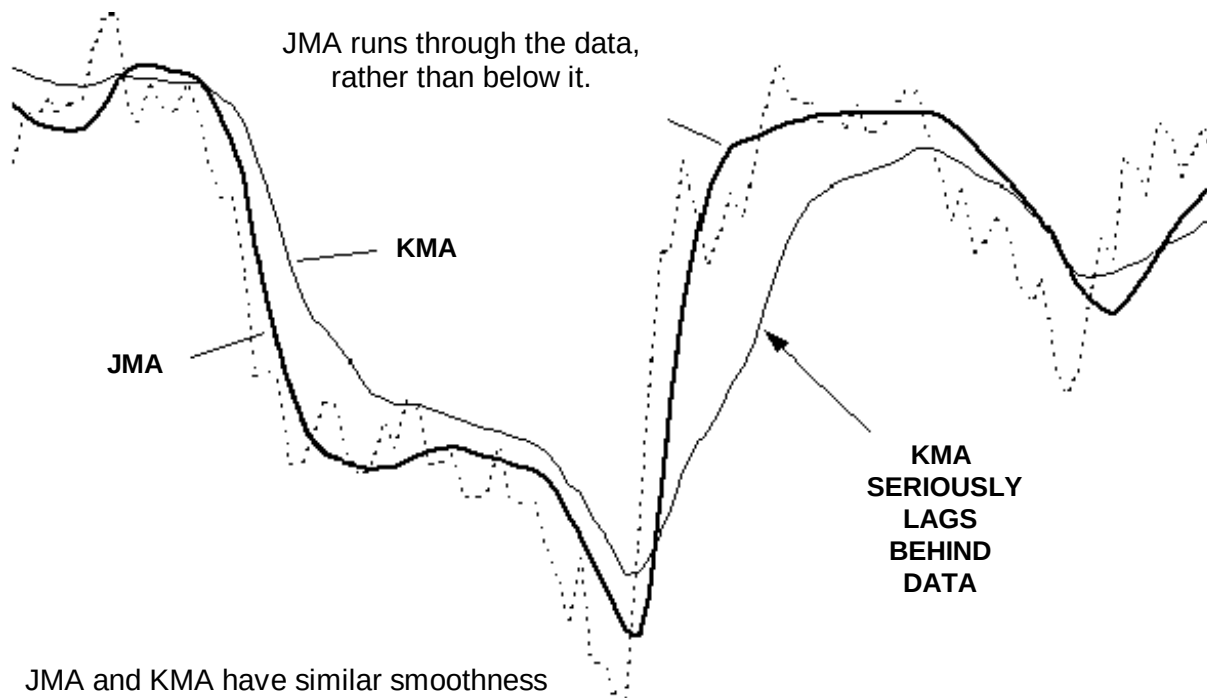
Most MA filters have another problem: they lag behind the original time series. This is a critical issue because excessive delay and late trades may reduce profits significantly.

Ideally, you would like a filtered signal to be both smooth and lag free. For many types of moving average filters, including the three classics (simple, weighted, and exponential), greater smoothness produces greater lag. For example, in the chart to the right, price action is the dotted line. The exponential moving average, EMA, lags well behind JMA (thick solid line). As you can see, with EMA's excessive lag, you would have had to wait a long time before it returned to the price action. In contrast, JMA never left it!



Adaptive filters developed by others, such as the Kaufman and Chande AMA, will also lag well behind your time series. Kaufman's Moving Average (KMA), is an exponential moving average whose speed is governed by the "efficiency" of price movement. For example, fast moving price with little retracement (a

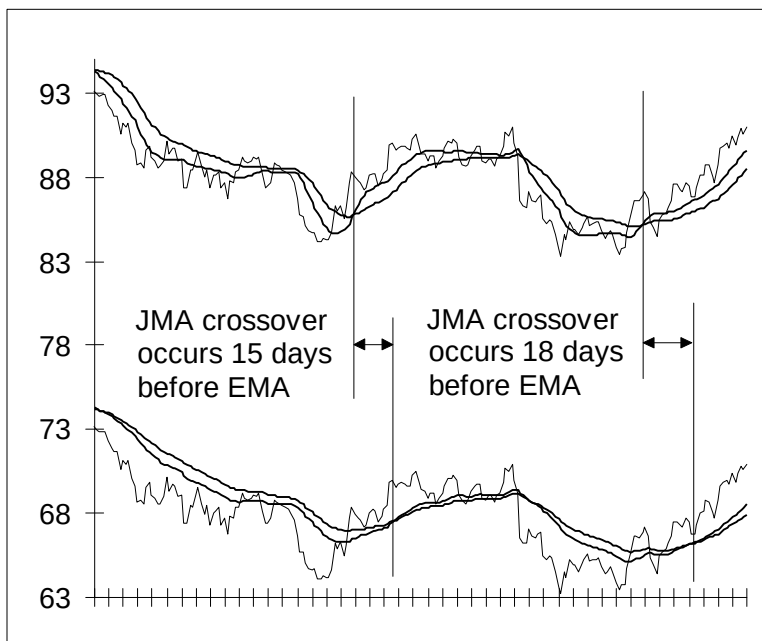
strong trend) is considered very efficient and the KMA will automatically speed up to prevent excessive lag. This interesting concept sometimes works well, sometimes not. For example, the chart below shows KMA lagging well behind JMA.



The advantage in avoiding lag is readily apparent in the chart to the right. Here we see how JMA enhances the timing of a simple crossover oscillator. The top half of the chart shows crude oil closing prices tracked by two JMA filters of different speed. The bottom half uses two EMA (exponential moving average) filters.

The oscillator becomes positive when the curve of the faster filter crosses over the slower one. This occurrence suggests a "buy" signal.

Note that JMA's crossovers are 15 and 18 days earlier! Can you afford to be 15 days late?

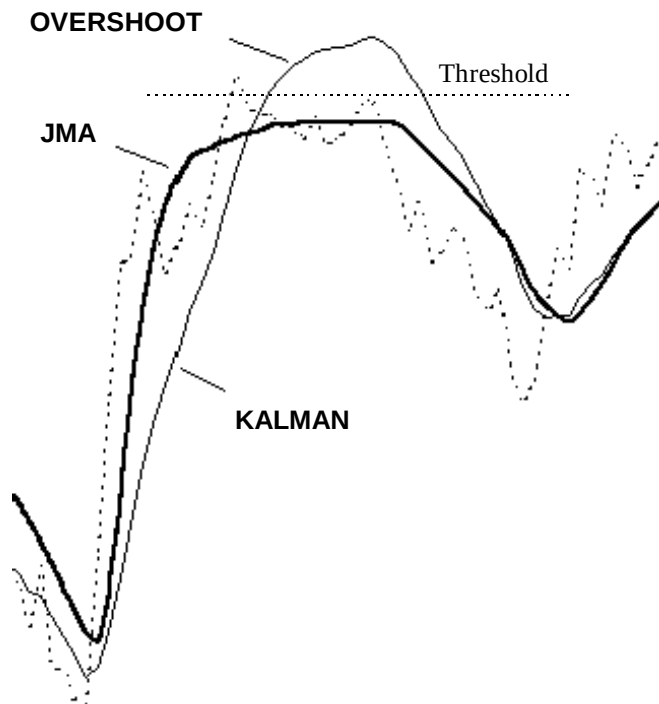


BENCHMARK #3: OVERSHOOT

Many trading systems set triggers to buy or sell when price reaches a certain threshold level. Because there is an inherent amount of noise in price action, the typical approach is to trigger when a moving average crosses the threshold. The smoothed line has less noise and is less likely to produce false alarms.

To do this right, you'll need an exceptional moving average indicator. Common versions lag too much and many sophisticated designs, like the Kalman or Butterworth filter, tend to overshoot during price reversals. Overshoots create false impressions of prices having reached levels it never truly did.

For example, in the chart to the right we see the famous Kalman filter overshoot price data, creating a false price level that the market never really achieved. DEMA filters also tend to overshoot. The overshoot crosses the shown threshold and triggers a false alarm. In contrast, **JMA did not overshoot** and thus avoided a false alarm with the user's set threshold.

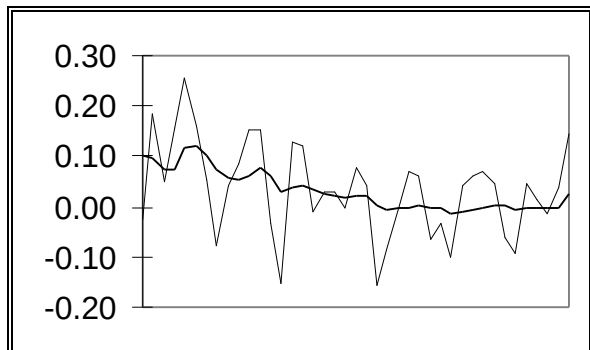
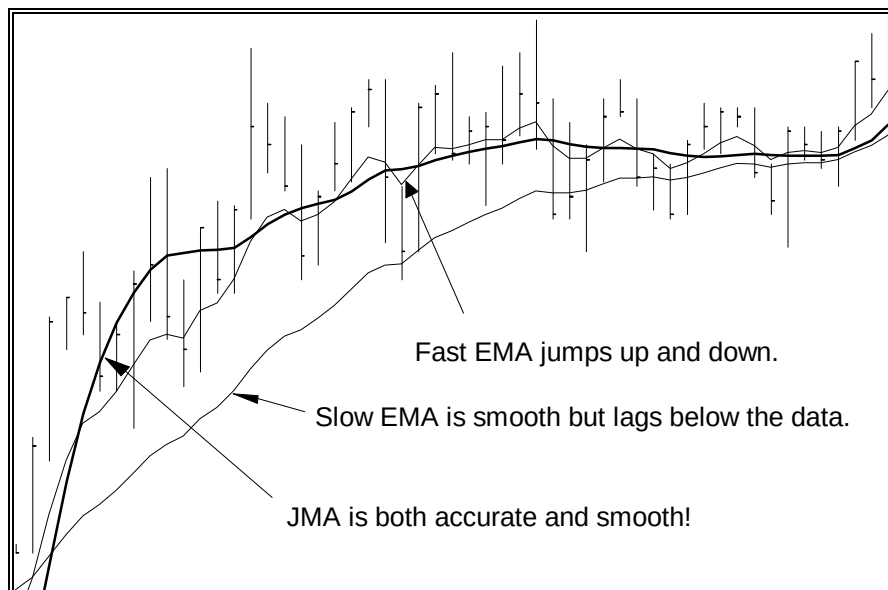


BENCHMARK #4: SMOOTHNESS

The most important property of a noise reduction filter is how well it removes noise, as measured by its smoothness.

In the chart to the right, EMA and JMA filters are run across closing prices. Note how much the fast EMA alternates upward and downward while JMA glides smoothly through the data. Clearly JMA reveals the noise-free underlying price more accurately.

If you try reducing EMA's erratic hopping by making it slower, you will discover its lag will become larger, producing late trade signals.

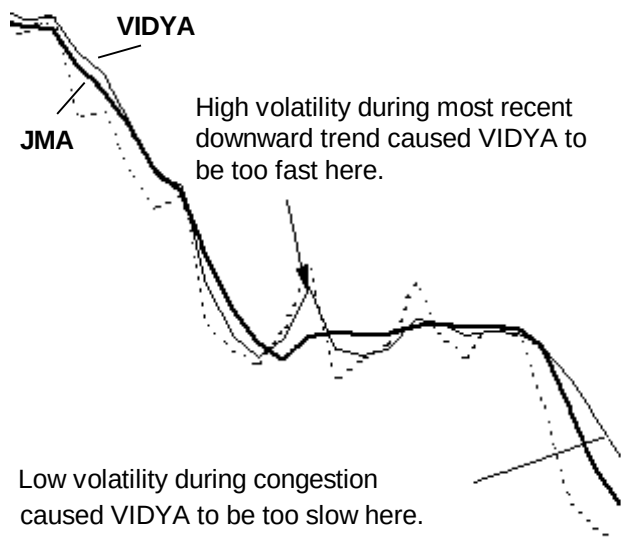


If you need a 2-bar momentum indicator, you could take the difference between two values along the EMA time series and produce the jagged line in the chart to the left. This is in contrast to the much smoother momentum signal based on JMA (flatter line). Imagine how many bad trades could be eliminated with this simple substitution!

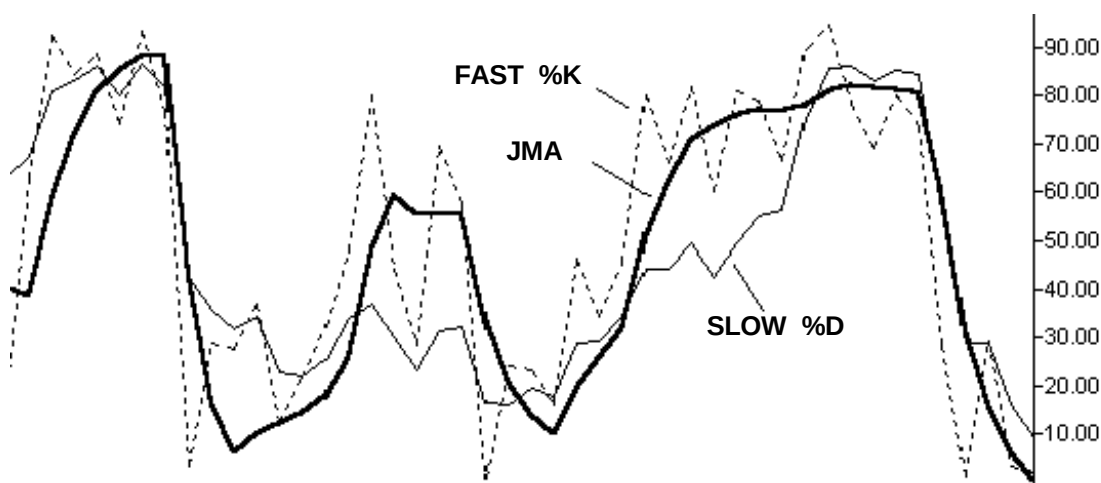
Moving averages should have consistent behavior. Some do not. For example, Chande's VIDYA is an exponential moving average whose speed is governed by the variance of price movement. Fast moving price has large variance which will eventually cause VIDYA to automatically speed up (in an attempt to prevent excessive lag). This concept sometimes works well, sometimes not.

In the chart to the right, JMA is the thick solid line and VIDYA is the thin solid line. Both perform approximately the same for the first 1/3 of the series. But due to the high volatility during this downward trend, VIDYA becomes hyperactive and fast tracks the choppy waves during the congestion phase of this time series. Smoothing is lost. In contrast, JMA cuts right through with a smooth horizontal line. In addition, the decrease in signal volatility soon causes VIDYA to slow down, too much in fact, as it lags behind JMA during the next downward price trend.

Comparing JMA to VIDYA



JMA also enhances the accuracy of technical indicators!

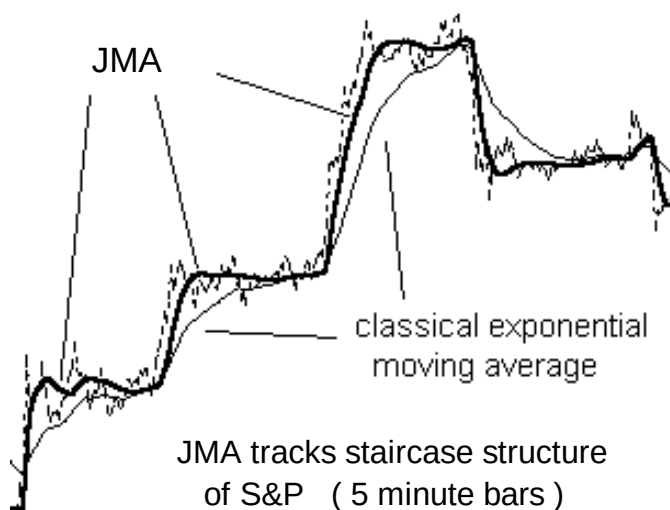


JMA resolves the riddle of how to get both smoothness and accuracy simultaneously, even with technical indicators! For example, the chart above compares the Fast %K indicator (dotted line) and two smoothed versions: one produced by the classic Slow %D (thin solid line) and the other produced by smoothing Fast %K with JMA (thick solid line). Clearly, JMA is both smoother and more accurate than slow %D!

JMA can also track price gaps produced by INTRA-DAY data. The chart on the right shows how JMA jumps to the next day's price levels while the classical exponential moving average lags behind.

Create superior trading indicators with ...

- **better timing**
- **less noise**
- **greater accuracy**



User Guide

for use in
Power Editor

After installing JMA, the indicator **JRC JMA** is ready for use. You may use it within TradeStation and SuperCharts the same way as you would use the exponential moving average indicator. The indicator **JRC JMA** consists of the following Easy Language code:

```
INPUTS: PRICE(CLOSE), LENGTH(8), PHASE(0) ;
VARS: JMAplot(0);
JMAplot = JRC.JMA(PRICE, LENGTH, PHASE);
PLOT1(JMAplot,"JRC JMA");
```

The first line of code says the indicator requires 3 input parameters. The first one is PRICE, which defines the time series to be smoothed. It defaults to the closing price of each bar. PRICE can be any simple calculation that produces a series, such as (High+Low+Close)/3, or any function that produces a series as its output, such as "RSI (close, 14)". In the later case, a line of code may look like this:

```
JMAplot = JRC.JMA(RSI(close,14), LENGTH, PHASE);
```

Although any type-series function can be used to generate input to **JRC.JMA**, we do not recommend using any time series other than simple combinations of Open, High, Low and Close, for example, (H+L+C) / 3. This gives results with the least lag.

The next parameter is LENGTH, and it determines the amount of smoothness to be applied to the time series. Larger values produce a smoother result. The default value for LENGTH is 8. The last parameter is PHASE, and it affects the filter's lag and can be either advanced (up to +100) or retarded (down to -100). It defaults to 0.

The third line of code calls the user function **JRC.JMA**. This user function contains a proprietary algorithm that performs the smoothing operation. It is encrypted and cannot be viewed. The last line tells the indicator to draw a plot of JMA's results.

Set the indicator's PROPERTIES so that its maximum number of bars referenced (**MaxBarsBack**) is as specified in the Executive Summary section of this manual.

You may want to design your own indicators that use the underlying user function directly. To see how to do this, read the "Executive Summary" earlier in this manual.

for use in
Quick Editor

SuperCharts users can easily build their own JMA indicator by specifying it as follows:

Indicator Name:	my_JMA								
Plot1 Formula:	JRC.JMA (price, length, phase)								
MaxBarsBack:	30 {see Executive summary for more info}								
Inputs:	<table><thead><tr><th>Name</th><th>Default Value</th></tr></thead><tbody><tr><td>Series</td><td>close</td></tr><tr><td>Length</td><td>8</td></tr><tr><td>Phase</td><td>0</td></tr></tbody></table>	Name	Default Value	Series	close	Length	8	Phase	0
Name	Default Value								
Series	close								
Length	8								
Phase	0								

The formula for plot1 may use complex expressions for PRICE. PRICE can be any calculation that produces a series, such as (High+Low+Close)/3, or any function that produces a series as its output, such as "RSI (close, 14)". In the later case, the formula for Plot1 might look like this....

```
JRC.JMA(RSI(close,14), LENGTH, PHASE);
```

Demonstrations

The remaining portion of this user manual contains demonstrations that show the power of JMA. The demonstrations include two simple systems for the S&P and two for US 30-Yr Bonds. You are invited to perform the same experiments as we did. Just follow the instructions in the manual, and you should get similar results.

First, you need to load in the **Bond data**, which is a text file of daily prices, from 1/84 - 1/90. The installer created two data files in the JMSOMEGA directory: USBONDS.TXT and SP500.TXT.

LOADING SAMPLE U.S. T-BOND DATA (or SP500 DATA) in TradeStation 4 / SuperCharts 4

1. Select menu command FILE / NEW WINDOW / CHART
2. In the INSERT PRICE DATA box select DIRECTORY radio button
3. Press NEW DIR and enter C:\JMSOMEGA\DATA\ in the DIRECTORY field and select ASCII
4. Press OK. Select USBONDS.TXT or SP500.TXT Select "Prompt for Format". Press PLOT
5. Select FIRST LINE OF DATA FILE and press OK.
6. In SETTINGS box
 - Select FUTURE as data type
 - enter "US" {for Bonds} or "SP" {for SP500} in the SEARCH FOR field
 - Deselect EXACT MATCH
 - Press FIND
 - Select "TREASURY BONDS 30 Yr" or "S&P500 Index" in the DATA NAME field
 - For the S&P, specify these price settings: Min Move = 5, Value = 500
 - Press OK
7. In FORMAT PRICE DATA box, select SETTINGS tab
8. For Bonds, enter 'First Date' as 1/03/84 and 'Last Date' as 1/03/90. For S&P, Enter 'First Date' as 1/01/83 and 'Last Date' as 1/10/96. Press OK

LOADING SAMPLE U.S. T-BOND DATA (or SP500 DATA) in ProSuite 2000 / TradeStation 2000

1. Select menu command FILE / NEW... / TradeStation Chart
2. Select menu command INSERT / SYMBOL ... , then press NEW DIR...
3. Select DATA TYPE as ASCII and press the BROWSE button and find the JRSOMEGA \ DATA folder
4. Press OK to exit the Browse box. Press OK to exit the NEW DIRECTORY box.
5. Select either USBONDS.TXT or SP500.TXT and press PLOT. Select "First Line of Data File"
6. Select date format MONTH/DAY/YEAR for SP500, or select YEAR/MONTH/DAY for USBONDS. Press OK
7. In the Settings box, set DATA TYPE: future
8. For S&P: set SEARCH FOR = SP.
For BONDS, set SEARCH FOR = US
9. Deselect "EXACT MATCH", press FIND
10. FOR S&P: Select "S&P500 Index" in the DATA NAME field, set MIN MOVE= 5, VALUE = 500. Press OK
FOR BONDS: Select "TREASURY BONDS 30 Yr", set MIN MOVE= 1, VALUE = 1000. Press OK
11. In Format Symbol box, set FIRST DATE = 01/01/83, LAST DATE = 01/10/96 Press OK

DEMO #1

Comparing Moving Averages

The best way to see how the Adaptive Moving Average works is to see some demonstration examples.

First, load some price data onto a chart. On the chart plot an exponential moving average indicator onto subgraph 1 (the price data) using the formula:

Xaverage(close,15)

Make sure its "maximum number of bars referenced" (MaxBarsBack) is set to 30, and its scale is set to "price data". Next plot the following indicator with a different color on the same chart:

JRC JMA(close,15,0)

Make sure its "maximum number of bars referenced" (MaxBarsBack) is set to 30, and its scale is set to "price data".

Note that although both have approximately the same smoothness, JMA tracks all the big moves more accurately than the exponential moving average. The LENGTH parameter determines the degree of smoothness and it can be any positive value. Small values make the moving average respond rapidly to price change and larger values produce smoother, flatter curves. Typical values for LENGTH range from 5 to 80. You can even use decimal numbers, such as 28.3.

DEMO #2

Controlling Lag with PHASE

Load any price data onto a chart. Create a "Custom 2 Line" indicator for subgraph 1 (the price data) and specify the formula for Plot #1 to be:

JRC.JMA (close, 15, -100)

and the formula for plot #2 to be:

JRC.JMA (close, 15, 100)

Make sure its "maximum number of bars referenced" (MaxBarsBack) is set to 30, and its scale is set to "Same as price data".

PHASE affects the amount of lag (delay). PHASE ranges from -100 (max lag) to +100 (min lag). Its default value is 0.

Note how the line with PHASE = -100 lags well behind the line with PHASE = 100. Although low lag is appealing, it is more likely to overshoot during price gaps than with lower values of PHASE. If price overshoot is not an issue, then you will probably prefer using positive values of PHASE.

If price overshoot is a real concern in your trading system, then you may prefer using negative values of PHASE. If you really don't care one way or the other, then leave PHASE at its default value of 0.

You can also optimize your system for the best value of PHASE.

DEMO #3

Threshold MACD Trading System

John Murphy, in his book, *Technical Analysis of the Futures Markets*, discusses the MACD indicator (by Gerald Appel). Typically, the MACD is simply the difference between the lines of two exponential moving average (EMA) filters with different speeds. Over time, the EMA lines are either converging (coming together) or diverging. Thus its name: Moving Average Convergence Divergence, or MACD.

With the MACD, a buy signal occurs when a faster moving average line crosses above a slower one and a sell signal occurs when the crossover is in the opposite direction. A simple trading rule using MACD might be: "If the MACD is positive (i.e., the faster line is higher than the slower one), be long (buy). If negative (i.e. the faster line is lower than the slower one), be short. (sell)."

Classical MACD indicators are great during trending price activity, riding the wave, so to speak. However, they are disastrous during choppy sideways activity, creating excessive, unprofitable trades.

This phenomenon occurs because moving averages lag behind the price signal and this lag causes a delay in transactions. During rapid price oscillations, this delay could be long enough to cause a sell trade to occur when the downward moving price has already hit bottom of a cycle or a buy trade to occur when the price has already reached the top of its cycle. Either way, the trader would experience a loss using classical MACD.

This demonstration shows how using JMA in a MACD system can improve the odds. JMA succeeds because you can control its lag. The installed trading system "**JRC thresh MACD**" was written using the QuickEditor so as to be readable in both SuperCharts and TradeStation. The system uses a signal generated by an installed user function called "**JRC.JMA.MACD**". This function takes the difference between two moving averages, just like a regular MACD, with the exception that the two moving averages are produced by JMA.

The system simply compares the MACD signal to two thresholds: Buyline and Sellline. If the signal crosses above buyline, then buy long. If the signal crosses below the sellline, then sell short.

The system, as viewed within the Power Editor, is as follows:

Input: series(close), L1(36), P1(40), L2(46), P2(-65), Buyline(-0.25), Sellline(0.65); IF JRC.JMA.MACD(series,L1,P1,L2,P2) crosses above BuyLine Then Buy on Close ; IF JRC.JMA.MACD(series,L1,P1,L2,P2) crosses below SellLine Then Sell on Close ;
--

Using the U.S. Bond data we supplied, insert the system "**JRC thresh MACD**", and set the following parameters:

Commission = \$30
Slippage = \$50
Margin = \$2700
Money Management (Stop Loss) = \$1550
Max number of bars referenced (MaxBarsBack) = 50
Default Trade Amount = 1 contract

NOTE: Before running any system, make sure Commission, Slippage and Margin are appropriate for the market being traded.

Set the following system input parameters:

series:	close	{ price series to be analyzed }
L1:	36	{ length of faster JMA }
P1:	40	{ phase of faster JMA }
L2:	46	{ length of slower JMA }
P2:	-65	{ phase of slower JMA }
Buyline	-0.25	{ threshold to enter long }
Sellline	0.65	{ threshold to enter short }

This very simple system was profitable during the 6 years of trading. \$46,000 profit is not bad for trading with \$2,700 margin. Note, however, that only 37 trades were taken and this may not be sufficient for assessing the value of this system.

This system is only for demonstration purposes. Do not trade real money using this system, it does not have all the necessary safety features for limiting exposure to risk.

JRC thresh MACD USBONDS.TXT-Daily 01/02/84 - 01/03/90

Performance Summary: All Trades

Total net profit	\$ 46602	Open position P/L	\$ 1343
Gross profit	\$ 75997	Gross loss	\$ -29395
Total # of trades	37	Percent profitable	49%
Ratio avg win/avg loss	2.73	Avg trade(win & loss)	\$ 1259
Max intraday drawdown	\$ -7925		
Profit factor	2.59	Max # contracts held	1
Account size required	\$ 10625	Return on account	439%

NOTE

Historical back-testing does not prove a system will be profitable in the future, but it can demonstrate whether or not a system would be worthless in the future.

The example trading systems described in this manual are for illustration purposes only. Do not trade real money using these demonstration systems.

A real trading system should be tested extensively for various kinds of flaws, including over sensitivity to parameter settings.

A real trading system requires not one but several mutually concurring indicators as well as good money management rules useful for assessing how much to invest and for placing exit stops.

DEMO #4

Reverse MACD Trading System

The classic MACD strategy does not work well in markets that frequently reverse rather than trend. In some markets, taking the opposite strategy (buy instead of sell and sell instead of buy) actually works. The S&P500 is one such market.

The installed trading system "**JRC reverse MACD**" was written using the QuickEditor so as to be readable in both SuperCharts and TradeStation. The system uses a signal generated by the same installed user function "**JRC.JMA.MACD**" as in demo #3.

The reverse MACD method is counter-intuitive. It trades the opposite of what the standard MACD would suggest. It compares the MACD signal to two thresholds: Buyline and Selline. If the signal crosses below the buyline, then buy long. If the signal crosses above the selline, then sell short.

This approach makes profits during price cycles, but loses during long trends. These losses can be limited with appropriate money management.

The simple system, as viewed within the Power Editor, is as follows:

```
Input: series(close), L1(46), P1(76), L2(74), P2(-60), Buyline(0.13), Selline(-0.15);

IF JRC.JMA.MACD(series,L1,P1,L2,P2) crosses below BuyLine
    Then Buy on Open ;
IF JRC.JMA.MACD(series,L1,P1,L2,P2) crosses above SellLine
    Then Sell on Open ;
```

Load the S&P 500 demo data now and set the date window from . After inserting the system "**JRC reverse MACD**", set

Commission = \$30
Slippage = \$150
Margin = \$15,000
Money Management (Stop Loss) = \$3400
Profit Target = \$3700
Max number of bars referenced (MaxBarsBack) = 30
Default trade amount = 1 contract

We set input parameter values as follows:

series:	close	{ price series to be analyzed }
L1:	46	{ length of faster JMA }
P1:	76	{ phase of faster JMA }
L2:	74	{ length of slower JMA }
P2:	-60	{ phase of slower JMA }
Buyline	0.13	{ threshold to enter long }
Selline	-0.15	{ threshold to enter short }

Even though this system has only 3 lines of code, it was profitable.

Total net profit	\$	95155
Total # of trades		84
Avg trade(win & loss)	\$	1132
Profit factor		2.00
Max intraday drawdown	\$	-13910
Return on account		329%

In a real trading system, max drawdown should be much smaller. Nonetheless, getting any profit out of the S&P is no easy task. Once again, we see the profit potential in using lag-controllable signals.

DEMO #5

Enhancing other Technical Indicators

Some indicators are inherently noisy, jumping up and down. One way to smooth them out is to increase their window length. For example, consider the 14-day RSI indicator. You can increase its length from 14 to 20 by using **RSI(close, 20)**. However, you still get a jagged curve as well as a decrease in the RSI's effective range.

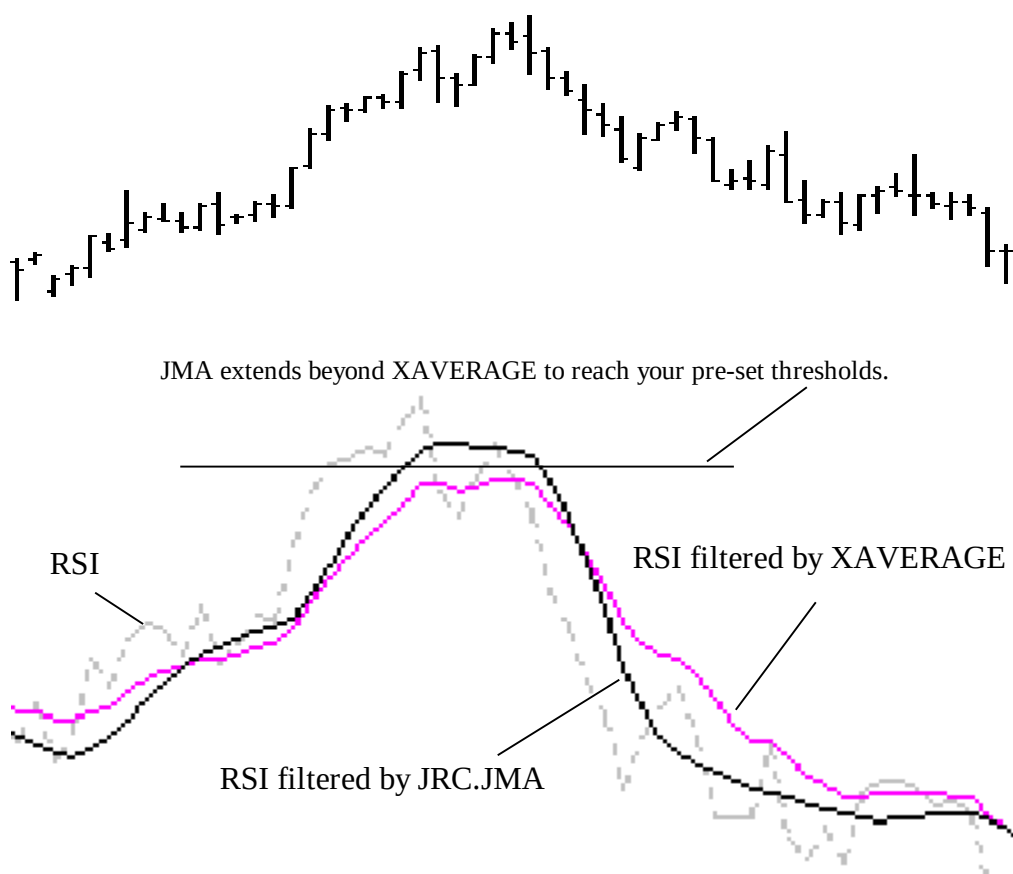
On the other hand you could run the original indicator through a smoothing filter, such as an exponential moving average by using

XAVERAGE(RSI(close,14), 10)

Although this reduces the jagged motion, it also reduces the RSI's effective range as shown below. A better way is to smooth the RSI with JMA by using

JRC.JMA (RSI(close,14), 10, 0)

The chart below illustrates how you get better smoothness using JMA as well as better dynamic range. The superior range improves chances that the signal will reach any particular threshold for determining oversold/overbought conditions. The superior smoothness improves the chances that your thresholds will be reached only when true conditions exist and not because of noise inherent in price charts and classical indicators.

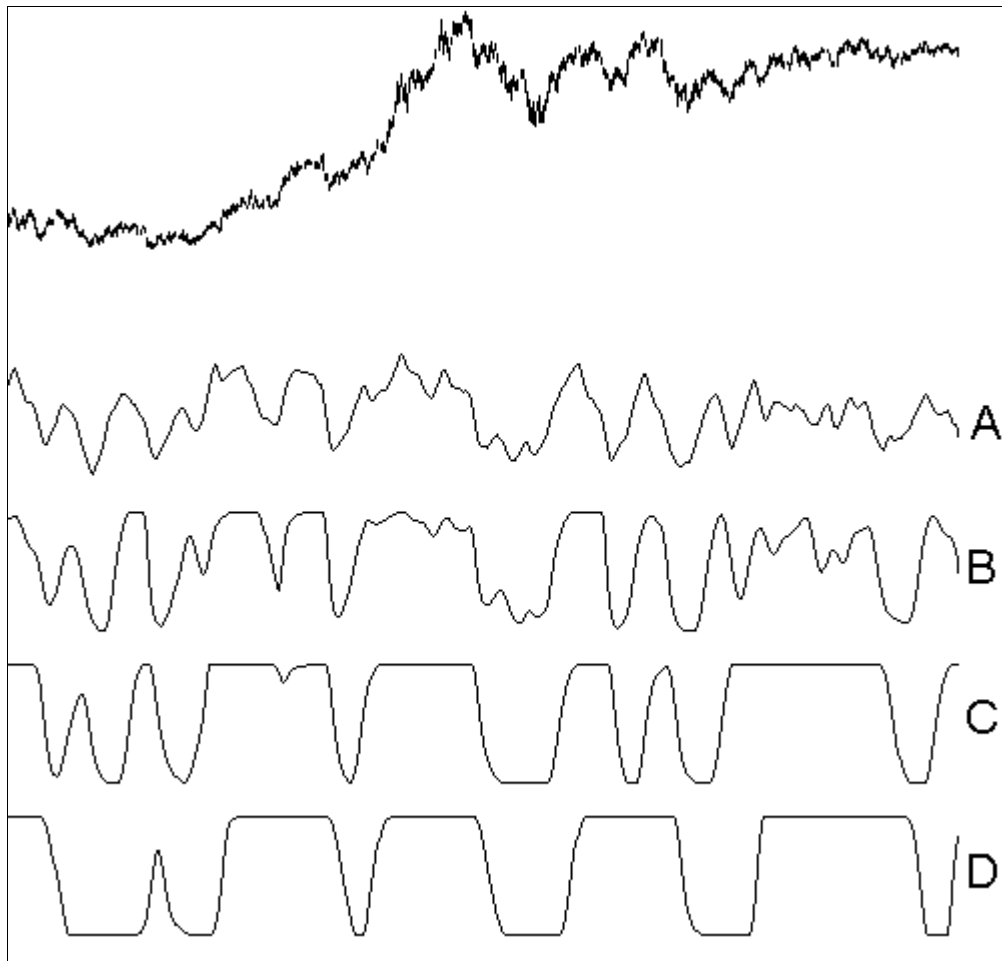


The following examples require Jurik tools JMA and RSX

This section illustrates the power of combining Jurik's **JMA** with another of our tools, the **RSX**. To run the demonstrations described herein, you will need to have installed both JMA and RSX onto your Omega Research application software. Details about RSX and how to acquire it are available on our web page at http://www.jurikres.com/catalog/ms_rsx.htm.

As shown earlier, classical indicators can be smoothed by applying JMA to the indicator's output. In addition, you can also apply JMA to data **before** it is fed to a classical indicator. This form of preprocessing may transform the nature of a classical indicator into a completely new function.

For example, plot **A** in the chart below shows Jurik's RSX indicator with no pre-smoothing by JMA. (RSX is Jurik's version of the classical RSI, only better.) In plots **B**, **C** and **D**, JMA's smoothness parameter is 10, 30 and 60 respectively. As pre-smoothness increases, note how RSX tends to yield more extreme values. In plot **D**, RSX yields only two values, its maximum and minimum, indicating "trend-up" and "trend-down".



The formula for plot A: JRC.RSX (JRC.JMA (close, 0 , 0) , 14)
 B: JRC.RSX (JRC.JMA (close, 10 , 0) , 14)
 C: JRC.RSX (JRC.JMA (close, 30 , 0) , 14)
 D: JRC.RSX (JRC.JMA (close, 60 , 0) , 14)

The next two demonstrations show how pre-smoothing with RSX yields trading systems better than those shown in demonstrations 3 and 4.

DEMO #6

Reverse RSX trading system

Demonstration #4 revealed how you can exploit the reversal nature of the S&P500 by waiting for the naturally lagging MACD to cross a threshold and then place a trade opposite to the way it is typically done. (i.e., long instead of short, and short instead of long). With proper money management, it makes a profit.

This demonstration proposes another system for trading reversal markets. It still uses the philosophy of Demo #4, but replaces the MACD with our own modified version of RSI, called RSX. The user functions are called **JRC.RSX** (type SERIES) and **JRC.RSX.flex** (type SIMPLE). The two functions produce the same values, and measure trend strength (not trend duration) in terms of what percentage of all current price action is in one direction. Just as with the classical RSI, if all price action is in the up direction (a strong uptrend), then its output is +100%. If all price action is in the down direction (a strong downtrend), then output is 0%. Sideways action produces a values close to 50%.

The installed trading system "**JRC reverse RSX**" was written using the QuickEditor so as to be readable in both SuperCharts and TradeStation. It uses a signal generated by the user function "**JRC.RSX**". The system's code contain the following three lines ...

```
Input: series(H+L), L1(5), P1(9), L2(22), lag(8) ;
IF JRC.RSX( JRC.JMA( series[lag], L1, P1 ), L2 ) crosses below 50
    Then Buy at market ;
IF JRC.RSX( JRC.JMA( series[lag], L1, P1 ), L2 ) crosses above 50
    Then Sell at market ;
```

For this experiment, we charted the same 13 years of daily S&P500 futures price data as in demo #4. After inserting the system "**JRC reverse RSX**", we set the following:

Commission = \$30
Money Management (Stop Loss) = \$3850
MaxBarsBack = 40
Slippage = \$150
Margin = \$15,000
Default Trade Amount = 1 contract

We set input parameter values as follows:

series:	H+L	{ price series to be analyzed }
L1:	5	{ length of JMA }
P1:	9	{ phase of JMA }
L2:	22	{ length of RSX }
lag:	8	{ added lag to the price time series }

This system is only for demonstration purposes. Do not trade real money using this system, it does not have all the necessary safety features for limiting exposure to risk. In addition, the maximum drawdown of this "paper traded" system may be too large for your trading account.

Once again, we see the profit potential in using lag-controllable signals. Even though this system has only 3 lines of code, it is more profitable than the MACD system in demo #4. The number of trades increased from 84 to 125 and returns increased from 329% to 510%. The more trades your system produces when paper trading, the more reliable are the performance results.

JRC reverse RSX	SP500.TXT-Daily	01/03/83 - 01/10/96	
Performance Summary: All Trades			
Total net profit	\$ 166250	Open position P/L	\$ 0
Gross profit	\$ 345485	Gross loss	\$-179235
Total # of trades	125	Percent profitable	58%
Ratio avg win/avg loss	1.37	Avg trade(win & loss)	\$ 1330
Max intraday drawdown	\$ -17580		
Profit factor	1.93	Max # contracts held	1
Account size required	\$ 32580	Return on account	510%

DEMO #7

Threshold RSX trading system

In this demonstration, we start with the same system as in the previous demo, but replace the reverse trading rule with the classical (non-reversed) version in order to make it useful in more trendy markets. In this application, we need to remove as much lag as possible. Consequently the LAG parameter was removed and replaced by two threshold values for triggering early buy/sell signals, as in demonstration #3.

The installed trading system "**JRC thresh RSX**" was written using the QuickEditor so as to be readable in both SuperCharts and TradeStation. The system uses a signal generated by an installed user function called "**JRC.RSX**". (Same as in demo #6).

The system simply compares the new signal to two thresholds: Buyline and Selline. If the signal crosses above buyline, then buy long. If the signal crosses below the selline, then sell short.

The system, as viewed within the Power Editor, is as follows:

```
Input: series(close), L1(2), P1(49), L2(9), Buyline(39), Selline(54) ;
IF JRC.RSX( JRC.JMA( series, L1, P1 ), L2 ) crosses above BuyLine
    Then Buy on Close ;
IF JRC.RSX( JRC.JMA( series, L1, P1 ), L2 ) crosses below SellLine
    Then Sell on Close ;
```

To run this experiment, load the **US Bond** data provided on the installation floppy disk. After inserting the system "**JRC thresh RSX**", set the following:

Commission = \$30
 Slippage = \$50
 Margin = \$2700
 Money Management = \$2550
 Max number of bars referenced (MaxBarsBack) = 50
 Default Trade Amount = 1 contract

Set the system input parameters as follows:

series:	close	{ price series to be analyzed }
L1:	2	{ length of JMA }
P1:	49	{ phase of JMA }
L2:	9	{ length of XMA }
Buyline	39	{ threshold to enter long }
Sellline	54	{ threshold to enter short }

This very simple Bond trading system was profitable during the 6 years of trading. It placed significantly more trades than **JRC thresh MACD** making it a more reliable study. With a \$2,700 margin and a max drawdown of \$9,400, the account size required for this study was approximately \$12,100. With a profit of \$60,000, that's a return of almost 500%.

This system is only for demonstration purposes. Do not trade real money using this system, it does not have all the necessary safety features for limiting exposure to risk.

JRC thresh RSX	USBONDS.TXT-Daily	01/02/84 - 01/03/90
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Performance Summary: All Trades

Total net profit	\$ 60361	Open position P/L	\$ 1093
Gross profit	\$ 119705	Gross loss	\$ -59343
Total # of trades	99	Percent profitable	49%
Ratio avg win/avg loss	2.06	Avg trade(win & loss)	\$ 609
Max intraday drawdown	\$ -9400		
Profit factor	2.02	Max # contracts held	1
Account size required	\$ 12100	Return on account	499%

NOTE

Historical back-testing does not prove a system will be profitable in the future, but it can demonstrate whether or not a system would be worthless in the future.

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A real trading system requires not one but several mutually concurring indicators as well as good money management rules useful for assessing how much to invest and for placing exit stops.

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